

**MATH 2211, SPRING 2026: MIDTERM 2**

This exam has 6 (six) printed pages. Please show all your work. Outside materials are NOT permitted for use on this test. Do not spend more than 10 minutes on any one part of a problem. If you get stuck for more than a few minutes, just move on to the next part. *You should feel free to use the results of earlier problems in your solutions to later ones, regardless of whether you have finished them or not.* You have 50 minutes to work on the test. Write your answers in the space provided.

Name \_\_\_\_\_

|           | Score |
|-----------|-------|
| Problem 1 | /10   |
| Problem 2 | /10   |
| Problem 3 | /10   |
| Problem 4 | /10   |
| Problem 5 | /10   |
| Total     | /50   |

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(1) (10 points) **True or false. Write one or two lines justifying your answer:**

(a) There is an invertible matrix  $A \in M_{n \times n}(F)$  with linearly dependent columns.

(b) There exists a linear transformation  $T : \text{Poly}_{\leq 5}(F) \rightarrow F^7$  with  $\dim \text{im } T = 4$  and  $\dim \ker T = 3$ .

(c) If a matrix  $A \in M_{n \times n}(F)$  is invertible, then so is  $A^T$ .

(d) Every linear transformation  $T : F^2 \rightarrow F^2$  is diagonalizable.

(e) If  $A \in M_{n \times n}(F)$  is invertible then  $\det(A^{-1}) = \det(A)^{-1}$ .

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(2) (10 points) Consider the linear transformation

$$T : \mathbb{Q}^3 \rightarrow \mathbb{Q}^3$$
$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \mapsto \begin{bmatrix} a - c \\ b + c \\ a + b \end{bmatrix}$$

Find bases for the kernel and image of  $T$ .

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- (3) (10 points) Suppose that  $(v_1, \dots, v_n)$  is a basis for  $F^n$ , and that  $A \in M_{n \times n}(F)$ . Show that the following are equivalent:
- (a)  $\det(A) \neq 0$
  - (b)  $(Av_1, \dots, Av_n)$  is also a basis for  $F^n$ .

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- (4) (10 points) Consider the matrix  $C = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} \in M_{2 \times 2}(\mathbb{Q})$  with associated linear transformation  $T : \mathbb{Q}^2 \rightarrow \mathbb{Q}^2$ .
- (a) Compute the characteristic polynomial and eigenvalues of  $C$ .

- (b) Show that  $T$  is diagonalizable, and use this to compute  $C^{10}$ .

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(5) (10 points) Let  $a, b, c$  be three *distinct* elements of a field  $F$ . Show that

$$\det \begin{pmatrix} \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix} \end{pmatrix} \neq 0$$